Jeffrey Kagle — Portfolio

Selected projects in acoustics, robotics, and product design.

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BIOMECHANICS · THEORY · DEC 2024 · EMAE 415 BIOMECHANICS · CASE WESTERN RESERVE UNIVERSITY · PRESENTED DECEMBER 2024

Biological and Robotic Synthesized Cuttlefish Fin Biomechanical Model

Jeffrey Kagle

Cuttlefish swim by passing a sinusoidal wave along muscular-hydrostat fins, but no experimental study had measured or modeled the thrust these fins produce for forward swimming. This theory and literature-review project synthesized biological fin mechanics (anatomy, EMG activation, and force–strain models) with a robotic sinusoidal-fin model into one hybrid force model, creating a benchmark for comparing cuttlefish and designing soft biomimetic robots.

↓ Final Paper

3D
fin force model: dorsoventral, lateromedial, and posteroanterior

2
fin beating modes captured: gentle (aerobic) and vigorous (anaerobic)

11
biological and robotic studies synthesized into one benchmark

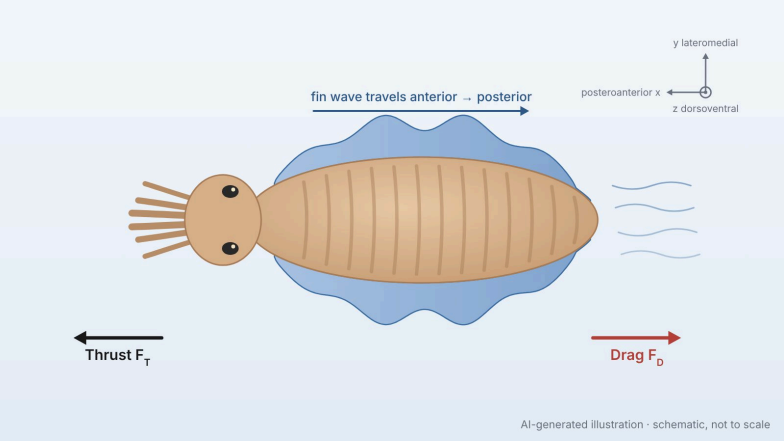


Fig. 1. AI-generated illustration (not from the paper): dorsal view of a cuttlefish with the fin wave travelling anterior to posterior, the resulting thrust and drag, and the body axes used in the model. Schematic, not to scale.

KEY CONTRIBUTIONS

- Built a hybrid thrust model by equating the robotic sinusoidal-fin equations of Su et al. with the biological fin-force model of Johnsen and Kier, filling the gap left by the lack of measured fin thrust data for cuttlefish.
- Linked fin physiology to propulsion: muscles fire in a sinusoidal activation wave, gentle beating uses only the aerobic transverse muscle, and vigorous beating recruits all three muscle groups for higher force.
- Included elastic energy stored and recovered by the muscular-hydrostat fin, and estimated energy output from work (E = F·d) since no intake-versus-output data exist for fin propulsion alone.
- Proposed validation experiments: a soft biomimetic cuttlefish robot in a flow tank with force transducers, PIV, and strain gauges, and a live-animal flow-tank test to measure fin thrust using Newton's second law.

THE HYBRID THRUST MODEL

No study has measured the thrust of a live cuttlefish fin, so the model borrows it from robotics. Assuming a robotic sinusoidal fin (Su et al., 2022) with the same size and shape as the cuttlefish fin, the robot's thrust and drag equations stand in for the animal's, while the biological model (Johnsen & Kier, 1993) supplies the out-of-plane fin forces.

1. Net force from motion. Over a short time segment Δt, the change in swimming velocity u gives the acceleration, and Newton's second law gives thrust minus drag, where m is the cuttlefish's wet mass:

$$F_T - F_D = m \frac{u(t+\Delta t) - u(t)}{\Delta t}$$

2. Drag at a point velocity. From the robotic model, drag is the viscous shear (fluid viscosity μ) integrated over the fin surface S:

$$F_D = \mu \int_S \left(\frac{\partial v}{\partial \delta} \cdot n_x \right)_{\delta = 0} dS$$

3. Point thrust. Combining the two isolates thrust, which must also equal the pressure p the beating fin exerts on the water in the swimming direction:

$$F_T(t) = \int_S p(t) \left(n_x \cdot dS \right) = m \frac{\Delta u}{\Delta t} + \mu \int_S \left(\frac{\partial v}{\partial \delta} \cdot n_x \right)_{\delta = 0} dS$$

4. Fin forces from the animal. The biological model gives the dorsoventral and lateromedial fin forces from tissue tension: T_c is the tension in the crossed connective-tissue fibres at angle θ, T_{dv} the

dorsoventral muscle tension, A_{tm} and s_{tm} the transverse-muscle area and stress, and each element follows $T = EA\varepsilon$:

$$F_v = 120\,T_c\cos\theta + 60\,T_{dv}, \qquad F_h = A_{tm}\,s_{tm}, \qquad T = EA\,\varepsilon$$

5. Combined. Together these give a three-dimensional fin force in the posteroanterior, lateromedial, and dorsoventral directions:

$$\mathbf{F}_{\mathrm{fin}} = (F_T,\ F_h,\ F_v)$$

SOFT ROBOTICS · 2025 · SENIOR PROJECT · CASE WESTERN RESERVE UNIVERSITY · ADVISOR: PROF. CHASE CAO

Liquid Dielectric Elastomer Actuator Motor for Underwater Propulsion

Jeffrey Kagle, Katharine Lobas, Jarod Lau, Yi Jin, Chase Cao

Rigid motors and hydraulics struggle under deep-water pressure and tend to be bulky and inefficient, while conventional dielectric elastomer actuators deliver high force but little displacement. We designed a deformable rotary motor driven by liquid dielectric elastomer actuators (Peano-HASELs), selected the concept through weighted trade studies, and built custom quarter-circle actuators and a 3D-printed motor housing as a proof of concept.

↓ Final Report ↓ Poster · CWRU Intersections 2025

4
liquid Peano-HASEL actuators pulling in sequence on one rotor

20 cm
motor diameter, 3D-printed frame on two ball bearings

5–9 kV
drive voltages tested; actuators deformed without damage

3.1
trade-study score for rotary motion (flapping 2.45, jetting 2.2)

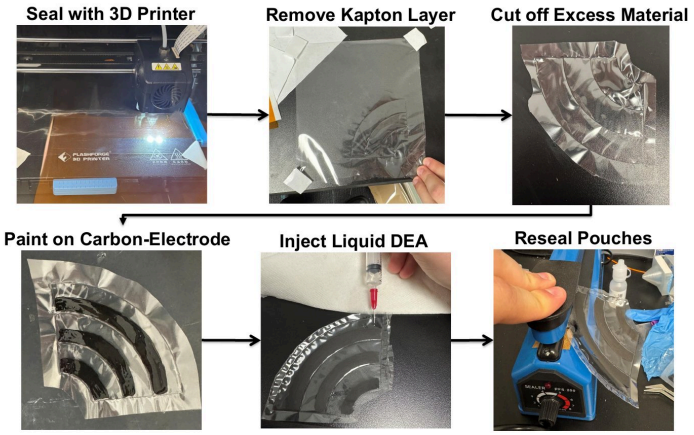


Fig. 1. Actuator fabrication: heat-seal BOPP film with a 3D printer, remove the Kapton layer, trim, paint carbon electrodes, inject the liquid dielectric, and reseal the pouches.

KEY RESULTS

- Weighted trade studies selected liquid DEAs as the actuation method (score 2.9, ahead of solid DEAs at 2.7, motors at 2.6, and hydraulics at 2.0) and rotary motion as the movement method (3.1 vs. 2.45 for flapping and 2.2 for jetting).
- Designed a three-pouch, quarter-circle Peano-HASEL actuator with painted carbon electrodes so four units fit around a circular frame and pull on an eccentric central rotor.

- Developed the fabrication process: heat-sealing BOPP film with a 3D printer, painting carbon electrodes, injecting silicone oil as the liquid dielectric, and resealing the pouches.
- Fabricated actuators deformed at 5, 8, and 9 kV from an off-board high-voltage DC/DC converter without damage; testing was limited to verifying functionality.
- Identified the main manufacturing fixes for future builds: thicker seal lines in the printer path to prevent leaks, better-placed injection points, and fewer air bubbles in the pouches.

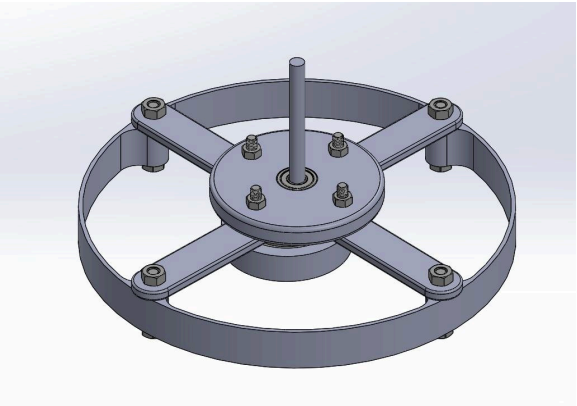


Fig. 2. Full CAD assembly: 3D-printed frame, support struts, and central hub with shaft on two ball bearings.

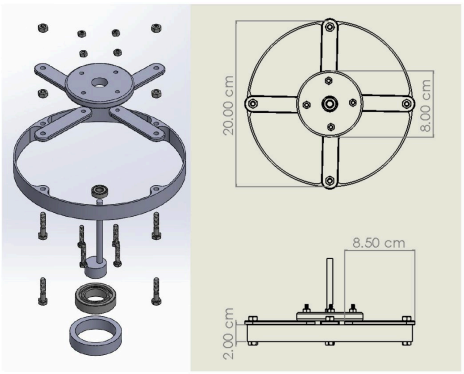


Fig. 3. Exploded view and dimensioned drawing of the 20 cm motor.

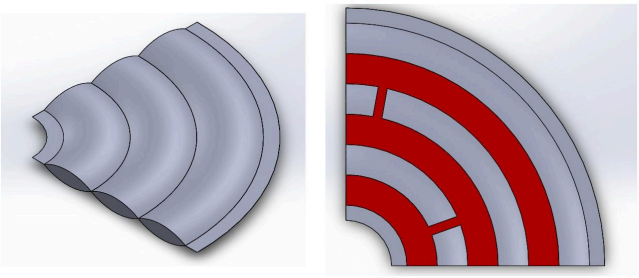


Fig. 4. Quarter-circle actuator design (left) and painted carbon electrode layout (right).

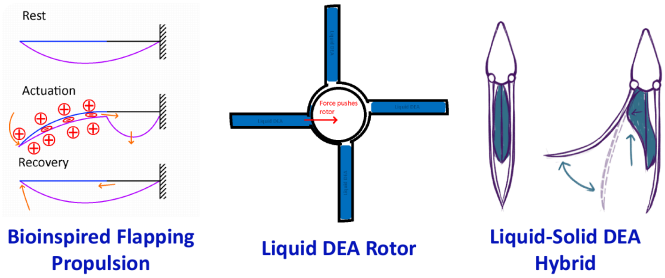


Fig. 5. Concepts compared in the trade studies: bioinspired flapping propulsion, liquid DEA rotor, and liquid–solid DEA hybrid.

PRODUCT DESIGN · 2026 · MAE 5630 ADVANCED PRODUCT DESIGN · CORNELL

ANCORE U Tiller Extension Joint

Mae Sliwinski, Garrett Smith, Alex Jenkins, **Jeffrey Kagle**

Competitive sailing hardware with a three-layer elastomeric architecture — polyurethane body, nylon rope core, and woven polymer mesh sleeve. Patent pending.

↓ Product Design Specification



Fig. 1. CAD renderings of the tiller extension (left) and the mesh-sleeved joint (right).